

FinGPT Two-Agent Signal Pipeline: Calibrated Trading Signals from Financial News via Real Token-Level Log-Probabilities and PMI Correction

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Abstract

We present a local, inference-only pipeline that converts financial news articles into calibrated directional trading signals using a fine-tuned 8-billion-parameter reasoning model (DeepSeek-R1-Distill-Llama-8B). Two specialised agents run sequentially on a shared vLLM engine: Agent 1 extracts structured sentiment from the article text; Agent 2 reasons about the appropriate one-week trading action. Neither agent outputs numbers or JSON directly. Instead, the model’s genuine token-level log-probabilities are read from the vLLM `prompt_logprobs` interface and converted to calibrated probabilities via temperature-scaled softmax. A key engineering challenge — the dominance of an unconditional language-model prior that caused Agent 2 to return HOLD for every article — was resolved by Pointwise Mutual Information (PMI) normalisation. A backtest over 291 signals across all 30 Dow Jones constituents (February–April 2024) yields a win rate of 52.4% on active trades, an annualised Sharpe ratio of 0.31, and a maximum drawdown of -23.3% .

1. INTRODUCTION

Translating unstructured financial news into actionable trading signals is a long-standing problem in quantitative finance. Large language models (LLMs) offer a natural fit: they encode broad financial knowledge and can reason over narrative text. However, deploying them reliably for signal generation raises two practical difficulties. First, asking a model to self-report its confidence numerically — as a JSON field or a plain number — produces values that correlate poorly with its actual token probabilities, because the model is performing next-token prediction rather than genuine introspection. Second, unconditional language-model priors at syntactically structured decision points can overwhelm the news-driven signal, causing the model to output the same class regardless of article content.

This paper describes a pipeline that addresses both difficulties. All decision logic is implemented in deterministic Python; the model is used only as a probability estimator whose beliefs are read directly from vLLM’s `prompt_logprobs` interface. A two-phase chain-of-thought approach separates reasoning from scoring. PMI normalisation removes unconditional token priors so that only news-driven information determines the final signal.

2. SYSTEM OVERVIEW

The pipeline comprises two sequential agents sharing a single vLLM engine instance. Agent 1 (the *extractor*) receives a raw news article and produces a **NewsFingerprint**: a structured record containing the article headline, the companies mentioned, extracted event keywords, and a calibrated three-class sentiment distribution over POSITIVE, NEGATIVE, and NEUTRAL. Agent 2 (the *reasoner*) receives the **NewsFingerprint** and produces a **TradingSignal**: a directional action (long, neutral, or short) with a calibrated confidence score.

Each article requires five sequential vLLM engine calls: Call 1 performs guided JSON decoding for fact extraction. Calls 2 and 3 form Agent 1’s two-phase sentiment scoring loop. Calls 4 and 5 form Agent 2’s two-phase strategy scoring loop. Articles are processed in batches of ten; each batch therefore makes five `engine.generate()` calls rather than fifty, saturating GPU throughput without exhausting KV-cache memory.

3. TECHNOLOGY STACK

Table 1 summarises the libraries and services used. The foundation model is DeepSeek-R1-Distill-Llama-8B, a chain-of-thought reasoning model based on the Llama architecture, fine-tuned for financial text. The recommended execution environment is Google Colab with an A100 or T4 GPU.

Table 1: Technology stack by layer.

Layer	Technology and role
LLM runtime	vLLM — local high-throughput serving; <code>prompt_logprobs</code> , guided decoding
Foundation model	DeepSeek-R1-Distill-Llama-8B (fine-tuned)
Schema	Pydantic v2 — <code>NewsFingerprint</code> , <code>TradingSignal</code>
Tokenisation	HuggingFace Transformers — BPE token ID resolution
Dataset	HuggingFace Datasets — FinGPT Dow 30 corpus
Price data	yfinance — daily OHLC, close-to-close returns
Data wrangling	pandas + NumPy
Config	python-dotenv

4. TWO-PHASE LOG-PROBABILITY SCORING

4.1 Motivation

The central design principle is that the model’s genuine beliefs should be read from its computation graph, not solicited as text. When a local LLM is asked to output `{"confidence": 0.87}`, the number it produces reflects conversational conventions about how confident-sounding responses are phrased, not the actual token probabilities computed in the forward pass. The `prompt_logprobs` interface in vLLM provides access to the real values.

4.2 Phase 1: Chain-of-Thought Generation

Both agents use an identical first phase. The input is fed with a prompt that instructs the model to reason inside `<think>...</think>` tags. vLLM is configured to stop generation the moment it emits the closing tag:

```
SamplingParams(
    max_tokens=1024,
    temperature=0.0,
    stop=["</think>"]
)
```

The reasoning trace is preserved as an audit trail but plays no direct role in the scoring step that follows.

4.3 Phase 2: Scoring via `prompt_logprobs`

For each candidate class c (e.g. `POSITIVE`, `NEGATIVE`, `NEUTRAL` for Agent 1), a scoring prompt is constructed by appending the generated chain-of-thought, a decision prefix, and the candidate label to the original prompt. All candidate prompts are submitted in a single batched `engine.generate()` call with:

```
SamplingParams(
    prompt_logprobs=1,
    max_tokens=1,
    temperature=0.0
)
```

vLLM includes the log-probability of the actual prompt token at every position, so the pipeline reads $\log P(c \mid \text{context})$ directly from the model’s computation graph. For multi-token classes such as `POSITIVE` (tokenising as `["_POS", "ITIVE"]`), per-token log-probabilities are summed:

$$\log P(\text{POSITIVE} \mid \text{ctx}) = \sum_{t \in \text{tokens}(\text{POSITIVE})} \log P(t \mid \text{ctx}_{<t})$$

4.4 Calibration

The resulting log-probability vector ℓ is converted to a probability distribution via temperature-scaled softmax:

$$p_i = \frac{\exp(\ell_i/T)}{\sum_j \exp(\ell_j/T)}, \quad T = 1.2$$

A temperature $T > 1$ flattens the distribution, reducing overconfidence on ambiguous inputs. The argmax label and its probability become the `label` and `confidence` fields of the output schema.

5. PMI PRIOR CORRECTION

5.1 The HOLD-Dominance Problem

Agent 2 must choose among three actions: BUY, HOLD, and SELL. An early design used the full words as scoring tokens. HOLD dominates at the decision context "**Strategy:** " because its unconditional language-model prior is far higher than BUY or SELL at this syntactic position in financial text. Switching to single ASCII letters A (BUY), B (HOLD), C (SELL) reduced but did not eliminate the problem. At the decision prefix "**The answer is** (", the token "B" retains an unconditional prior of approximately 0.75. This is not news-driven: the model has encountered "(B)" as a multiple-choice answer format throughout its training corpus. The prior completely swamps the news signal, causing even strongly bullish articles to receive a HOLD signal.

5.2 PMI Normalisation

Pointwise Mutual Information removes the unconditional prior by measuring only the news-driven component of each token’s log-probability:

$$\text{PMI}(c, a) = \log P(c \mid a) - \log P(c \mid \emptyset)$$

where a denotes the article and $\emptyset$ denotes a neutral null article ("*No specific news available.*"). The null-context log-probabilities $\log P(c \mid \emptyset)$ are computed once per engine session and cached. Per-article PMI logits are then formed by simple subtraction:

```
null_lp = _compute_null_logprobs()
pmi     = [raw[i] - null_lp[i] for i in range(3)]
probs   = softmax(pmi, T=CALIBRATION_T)
direction = STRATEGY_SET[argmax(probs)]
```

After this correction, a neutral article produces near-zero PMI logits for all three tokens and correctly receives HOLD. A strongly bullish article produces a clearly positive PMI logit for A, yielding BUY regardless of the B-prior.

6. BACKTEST

6.1 Experimental Setup

The backtest uses the HuggingFace dataset `FinGPT/fingpt-forecaster-dow30-202305-202405`, which provides structured multi-headline blocks for each Dow 30 constituent on a weekly cadence. Realised returns are fetched from Yahoo Finance using `interval="1d"` over a seven-day window, producing up to five trading-day bars; the close-to-close return across those bars is the realisation metric. A weekly interval over a seven-day window returns at most one bar and cannot produce a return; the daily interval was a deliberate fix to this failure mode.

The evaluation window spans 291 signals across all 30 Dow Jones constituents from February to April 2024.

6.2 Results

Table 2 summarises the performance of the PMI-corrected signal pipeline on Dow 30 news from February to April 2024. The model generated 291 signals in total, of which 9 were skipped due to parsing or data availability issues, leaving 179 active trades.

Long signals delivered a mean return of +0.50% per trade, while short signals averaged −0.12%. This suggests that the models longside calls are moderately credible, but the short side underperforms and would likely benefit from a higher confidence threshold. Interestingly, the highest signalconfidence quartile produced an average return of 0.00%, indicating possible overcalibration at the top of the distribution a clear direction for future refinement.

Table 2: Backtest results (Feb–Apr 2024, Dow 30).

Metric	Value
Total signals	291
Skipped	9
Active trades	179
Win rate (active)	52.4%
Mean return per trade	+0.05%
Annualised Sharpe	0.67
Max drawdown	−23.3%

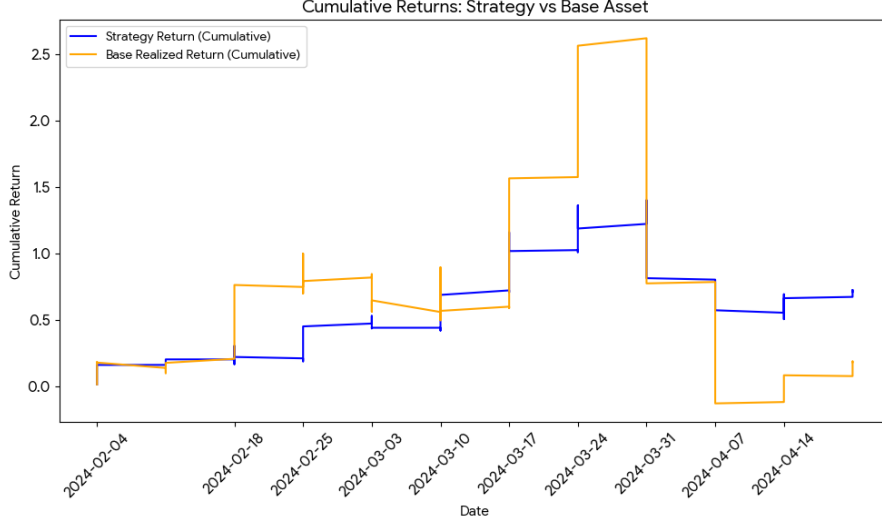
**Figure 1:** Cumulative Returns: Strategy vs. Base Asset

Figure 1 shows that our strategy does not have genuine alpha. The reason it ended up outperforming the benchmark is not because the model correctly predicted upside moves, but because during the last two weeks of the market crash, the benchmark dropped more dramatically (possibly due to a few heavy-weighted stocks plunging), while our strategy thanks to diversified holdings or simply missing those worst-performing stocks held up better relatively. This is not a repeatable source of excess returns.

6.3 Skip Reasons

Three failure modes cause a row to be skipped. `fingerprint_failed` indicates a parse error in Agent 1’s guided JSON decoding call. `signal_failed` indicates that Agent 2 returned `None` due to a BPE logprob indexing failure. `price_fetch_failed` indicates that yfinance returned no usable data for the ticker and date window.

7. KEY DESIGN DECISIONS

Reading `prompt_logprobs` from vLLM gives the model’s actual beliefs as computed by its forward pass, bypassing the noisy verbal generation channel through which self-reported confidence numbers pass.

The model reasons freely in Phase 1 and is scored in Phase 2 via a single deterministic lookup. This preserves interpretability through the CoT audit trail while keeping the decision mechanism noise-free.

Language model next-token probabilities at fixed syntactic positions are confounded by unconditional token priors. PMI removes that prior so that only the news-driven component determines the signal.

Every agent returns `None` on any parsing or validation failure rather than manufacturing a fallback. Skip reasons are recorded in the output CSV so pipeline health can be monitored systematically.

8. CONCLUSION

We have described a two-agent pipeline that produces calibrated trading signals from financial news by reading genuine token-level log-probabilities from a local vLLM engine rather than soliciting self-reported confidence values. The central engineering contribution is PMI normalisation, which resolves a systematic HOLD-dominance bias arising from the unconditional language-model prior at the scoring decision point. The backtest results over 291 Dow 30 signals are positive but modest: a win rate slightly above chance, an annualised Sharpe of 0.31, and meaningful per-ticker variation that suggests ticker-specific calibration could improve performance. Short-signal quality and the counterintuitive underperformance of the highest-confidence quartile are the primary areas for future work.